Proposal

Changes to "How a run is scored" (`sec:pf:measures`)

For review before anything is applied

Version 2, 5 October 2026

This document collects every change proposed for the scoring section of `Chapters/2.Introduction.tex`, so they can be reviewed together before any of them is applied.

Current state. The readability rewrite from earlier in this session is in the working tree of `thesis/`. It is uncommitted and changes only wording and order, not the model. Nothing else in this document has been applied. The two figures shown here are drafts, saved only in `scoring_proposal/` at the workspace root and not in `thesis/Images/`.

What changed in version 2. Your answers are worked in. The congestion part now follows your proposal and has no options left. Robustness (C) and unfinished tasks (D) are explained with worked examples, since you asked for more on both. Green boxes mark what is agreed, and orange boxes mark what is still open.

Item	Where	Status
Fixes P1 to P11 that need no model decision	Section 1	please confirm
A. What counts as even traffic	Section 2	open
B. Congestion as the mean of $\delta_i(t)$	Section 3	your proposal, please confirm
C. Robustness	Section 4	open, explained further
D. Blocked time of unfinished tasks	Section 5	open, explained further
E. Keep "movement cost", add one clause	Section 6	agreed (E1)
New structure of the section	Section 7	agreed
New concentration figure	Section 8	agreed
Timeline figure	Section 8	agreed, one point to confirm

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1 Check against the rest of the thesis

The section was compared with the Project Description (independent and dependent variables, research questions), the task lifecycle, the controller and congestion subsections, the scope subsection, the notation table, and the Method chapter (reward, conflict resolution, instrumentation, computational cost).

1.1 What already matches

- The seven dependent variables of the Project Description (service time, throughput, movement cost, waiting, concentration, backlog, runtime) are the seven quantities the section defines.
- The factors in `eq:rqformal` (window, congestion sensitivity, load m and λ_{task}, conditional communication) are the independent variables of the Project Description.
- τ_1 in the toy table (released 5, assigned 9, finished 14) matches the task example of `fig:taskroute` and `eq:lifecycle`.
- $\omega_j(t)$ comes from the override flag of the resolution operator, used the same way in the reward and in the instrumentation section.
- The notation table entries for Λ_T, $\bar{\zeta}_T$, μ_T, H_T, C_T, B_T, ω_j, W_T, κ and the weights $q_\bullet$ agree with the section.

1.2 What does not match

The problems are numbered P1 to P11, so they are not confused with the options of the decisions below.

	Mismatch	Proposed fix
P1	$\mu_T(e)$ is counted "over the evaluation window". Everywhere else "window" means what a controller may condition on (`tab:information`). The notation table repeats the phrase.	Say "over the horizon T" in the section and in `2.Notation.tex`.
P2	The section says how $\kappa(t)$ is measured "is left to the implementation". `sec:impl:cost` already fixes it: assignment and routing logged separately, same hardware, training and inference reported apart.	Point to `sec:impl:cost` and state that $\kappa(t)$ is the sum of the assignment and routing times.
P3	`eq:throughput` writes out $t_j^{\text{finish}} - t_j^{\text{released}}$, although ζ_j is already defined in `eq:servicetime`.	Write ζ_j and cite `eq:servicetime`.
P4	The text says "congestion and concentration are reported as separate outcomes" and the main RQ names congestion, but no congestion outcome is defined.	Decision B, Section 3.
P5	The main RQ names "robustness". Nothing in the section measures it.	Decision C, Section 4.
P6	The Project Description calls W_T "waiting caused by contention". W_T also counts steps where the agent chose to wait with nobody in the way.	Change the Project Description to "time blocked after assignment".
P7	W_T sums only over completed tasks. An agent stuck until the end never completes, so its blocked time never counts, and a deadlocked run can show a low W_T.	Decision D, Section 5.
P8	"Movement cost" $\bar{c}_T$ sums $\hat{c}$, which includes the waiting cost c_{wait} (`eq:onestepcost`).	Decision E, Section 6.
P9	$\mathcal{J}$ takes expectations $\mathbb{E}[\cdot]$ without saying over what. The Method seeds the order stream, the policy and the environment separately.	One sentence saying the expectation is over the order stream and the random seeds, estimated by the mean over evaluation seeds.
P10	The constraints paragraph (`[G9]`) explains four of the five constraints in `eq:decisionproblem`. `eq:transition` is never mentioned, although the action space enforces it for every π as well.	Add `eq:transition` to the group that holds by construction.
P11	The instrumentation section lists the logging of H_T, ζ_j, W_T and B_T, but not of $\bar{c}_T$.	One clause in `sec:impl:logging`.

Outside this section and not proposed for change now, the Background chapter (`3.Background.tex`, around line 343) still says "information regimes".

2 Decision A. What counts as "even" traffic

2.1 Your question: how does the measure know it is a corridor?

It does not know. The measure never looks at geometry. It only sees how many times each edge was traversed. In this warehouse there is no open space either. Every vertex is a position in an aisle and every edge is a piece of aisle, so a "corridor" is simply a chain of edges, like the eight edges from a_4 to D1 in Figure 1.

The real question is what "even" is compared against. The current definition divides by $\log |E_T^+|$, the number of edges that were used. So it asks whether traffic is even among the edges the agents happened to use. If all agents drive only along those eight edges, and evenly along them, the answer is yes, and the run gets $H_T = 1$ and $C_T = 0$. That is the score for perfectly spread traffic, although the whole fleet is in one aisle. The bottom panel of Figure 1 shows this case.

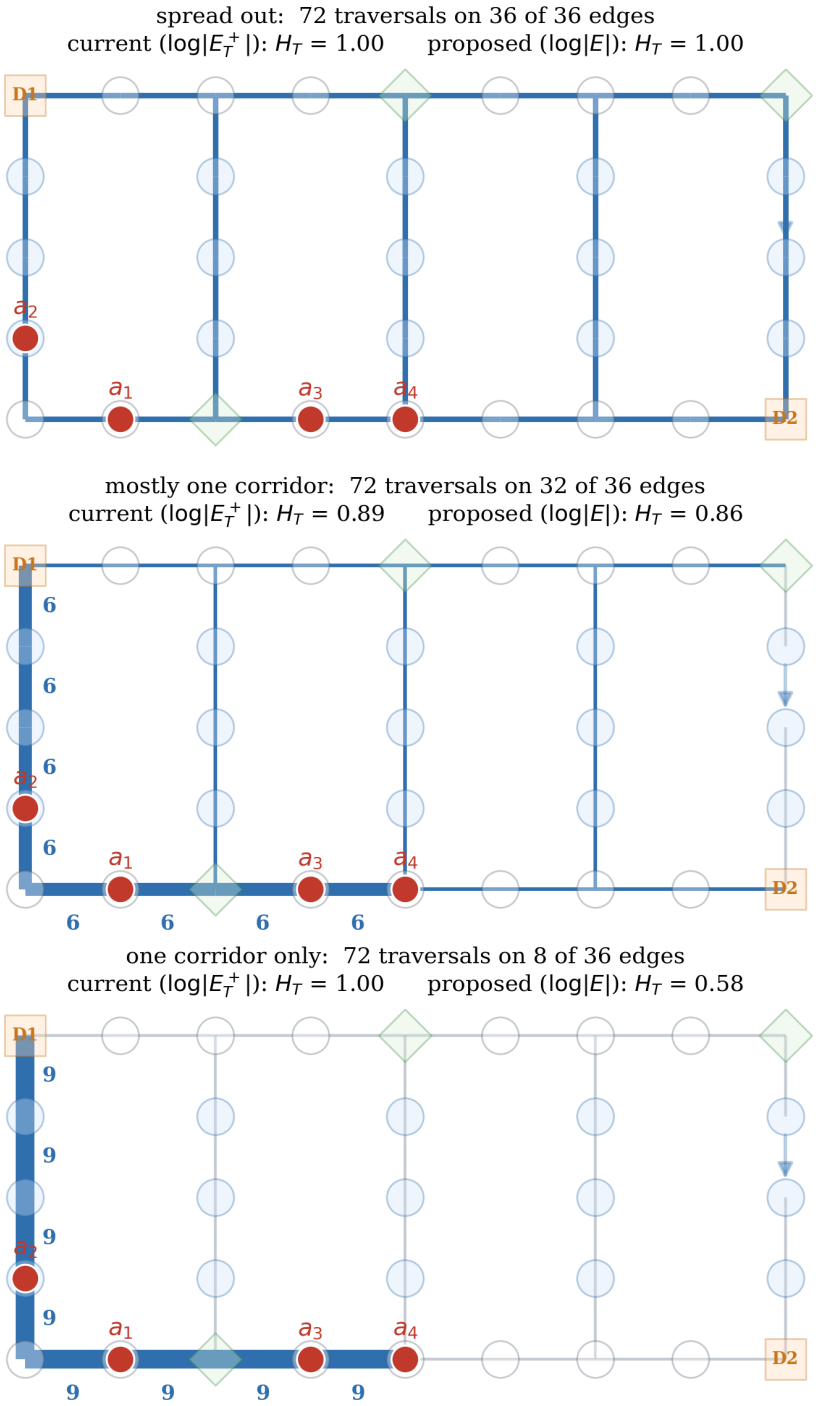


Figure 1: Draft of the new concentration figure, on the warehouse and agents of `observation.png`. Each panel has 72 traversals. The numbers are edge counts, and both normalisations are shown for comparison.

2.2 Options

A1. Divide by $\log|E|$, every edge in the graph. "Even" then means even over the whole warehouse. One corridor only gives $H_T = 0.58$. The cost is that $H_T = 1$ is out of reach in practice, because some edges are never needed. That does not harm the study. The graph is the same in every condition, so the constant is the same for every run, and a comparison of F_{fix} with an adaptive F stays fair. It also removes the stated limit that runs using different

numbers of edges are not comparable.

A2. Keep $\log|E_T^+|$. State the corridor case as a limit. C_T then misses exactly the failure the RQ asks about, which is feedback piling traffic onto a few aisles.

A3. Report a simpler share as well. The share of all traversals carried by the busiest 10% of edges (4 of 36). This gives 11%, 33% and 50% for the three panels. A reader understands it at once, but it is not smooth enough to weight in $\mathcal{J}$.

Recommendation. A1 for C_T in $\mathcal{J}$. Optionally add A3 as a reported quantity, not weighted. A1 is a model change, logged as a new `[Axx]` tag with a `GAPS.md` entry.

Your decision. Still open. A1, A2, or A1 with A3? The concentration figure is agreed either way. Only its numbers and the number of panels depend on this choice.

3 Decision B. Measuring congestion with the congestion feature

3.1 What you proposed

Measure congestion with the same per-agent crowding reading that the controllers already use, $\delta_i(t)$ (`eq:congestion`), the fraction of an agent's field of view occupied by other agents. Version 1 offered this as option B2 next to the alternative of not measuring crowding at all. You chose it, so the options are gone, and the version 1 subsection with the options is not needed any more.

3.2 How it becomes a score

$\delta_i(t)$ is one number per agent and per timestep. A run needs one number in total, so the readings are averaged over every agent and every timestep of the horizon,

$$\bar{\delta}_T = \frac{1}{mT} \sum_{t=0}^{T-1} \sum_{i=1}^{m} \delta_i(t).$$

For example, at $t = 11$ in `observation.png`, agent a_1 has $\delta_1 = 2/5 = 0.4$. If the four agents read 0.4, 0.2, 0.4 and 0.2 at every timestep, the run gets $\bar{\delta}_T = 0.3$, meaning that on average 30% of the space around an agent was taken.

The reading is defined the same way for every agent and every architecture (`fig:congestion`). So $\bar{\delta}_T$ compares the three controllers on equal terms, which is the reason the feature was defined that way in the first place.

3.3 How this relates to CRAMP

CRAMP (Pham et al. 2024, arXiv 2309.10275, the paper behind the crowding feature) computes the same kind of density for each agent, the number of agents in its field of view divided by the free space there. It uses the density only inside its reward. An agent is penalised for moving into an area whose density is above a threshold and rewarded for moving out of one. CRAMP does not report the density as a result. Its results are success rate, makespan, total moves and collision count (Section V-B of the paper).

So the thesis would use the CRAMP-style per-agent density in two places, as the input that congestion-sensitive routing reacts to, which it already does, and averaged into $\bar{\delta}_T$ as a reported outcome, which is new. A CRAMP-style count of how often an agent stands in a crowded area would need a threshold value that the thesis does not fix, so the plain average is the simpler choice.

3.4 Why W_T stays as well

$\bar{\delta}_T$ says how crowded it was. W_T says how much time the crowding cost. Agents moving in a tight convoy that never stops have a high $\bar{\delta}_T$ and a W_T of zero. The study needs both readings to answer RQ3. When congestion sensitivity is switched on, $\bar{\delta}_T$ shows whether crowding went down, and W_T shows whether that saved any time.

$\bar{\delta}_T$ is reported and not weighted in $\mathcal{J}$. Its cost already enters $\mathcal{J}$ through W_T, so weighting both would count one effect twice.

3.5 What changes in the thesis

- One short paragraph and the formula for $\bar{\delta}_T$ in the waiting group of the scoring section. It points to `fig:congestion`, so no new picture is needed.
- The sentence "congestion and concentration are reported as separate outcomes" then names $\bar{\delta}_T$ and W_T.
- A row in the summary table (not in $\mathcal{J}$), a row in the notation table, an `[Axx]` tag and a `GAPS.md` entry.
- "Crowding" is added to the dependent variables in the Project Description.

Your decision. Confirm that $\bar{\delta}_T$ is added this way, reported and not weighted in $\mathcal{J}$.

4 Decision C. What "robustness" means

4.1 The problem

The main research question asks how coupling affects "efficiency, congestion, traffic concentration, and robustness". Each of the first three has its own quantity in the scoring section. Robustness has none, and the thesis never says what it means by it.

4.2 What robustness would mean here

A gain is robust when it does not depend on luck or on one particular setting. Two things can break a gain in this study. One is the random seed, which decides the order stream and the training run. The other is the load, the number of agents and how fast tasks arrive. The worked example below shows the gain of adaptive storage over fixed storage, $\mathcal{J}(F_{\text{fix}}, \pi) - \mathcal{J}(F, \pi)$ as a percentage, over five seeds at two loads. The numbers are invented for illustration.

Load	seed 1	seed 2	seed 3	seed 4	seed 5	mean	seeds where F wins
$m = 4$	$+12$	$+10$	$+11$	$+9$	$+13$	$+11$	5 of 5
$m = 12$	$+25$	$+30$	-5	$+28$	-8	$+14$	3 of 5

At $m = 12$ the mean gain is larger, but in two of five seeds adaptive storage is worse than fixed storage. The gain at $m = 4$ is smaller and robust. The gain at $m = 12$ is larger and not robust. A report of the means alone would rank them the wrong way round.

4.3 The proposal

No new formula. The scoring section gets two sentences that say robustness is read from numbers the study already produces:

- the spread of each quantity across seeds, as the smallest and largest value, or as the number of seeds in which adaptive storage wins, as in the table above
- whether the gain keeps its sign as the load grows (RQ4), and whether the backlog B_T stays bounded at high load

This matches `eq:rqformal`, which already asks for a margin "exceeding seed variance". The proposal gives that phrase the name the RQ uses.

The alternative is to remove "robustness" from the main RQ. That is simpler, but it drops a property the thesis does test.

Your decision. C1, define robustness in two sentences as above, or C2, remove the word from the RQ?

5 Decision D. Blocked time of tasks that never finish

5.1 The problem with a worked example

W_T averages the blocked time over completed tasks only. Take the three tasks of the timeline figure and add a fourth, τ_4. It is released at 11, assigned at 12, gets stuck at 14, and is still stuck when the run ends at $T = 20$ (Figure 2).

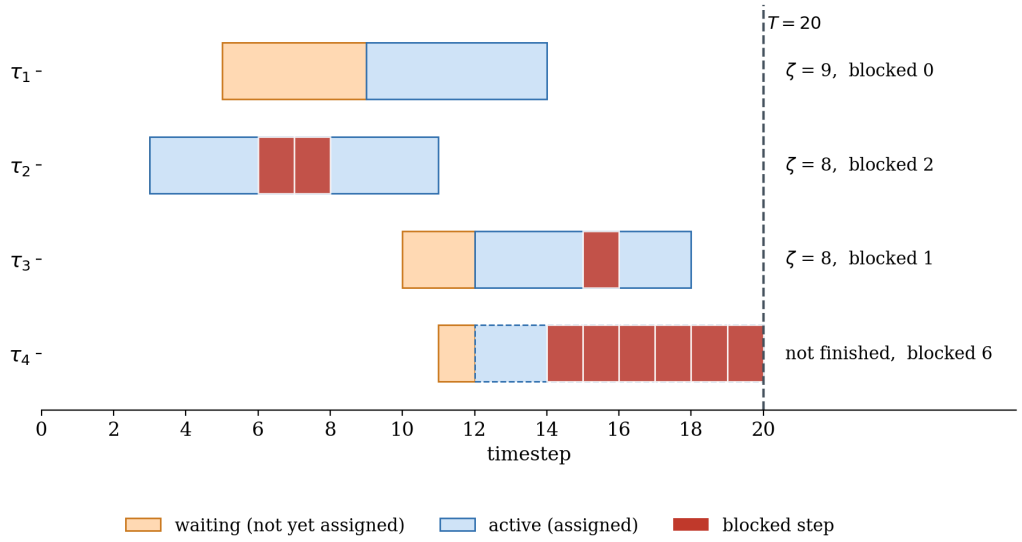


Figure 2: The timeline example with a fourth task τ_4 that is stuck from $t = 14$ and never finishes.

The two options give these numbers for this run.

	D1, completed tasks only	D2, every assigned task
W_T	$(0+2+1)/3 = 1$	$(0+2+1+6)/4 = 2.25$
$\bar{\zeta}_T$	8.33	8.33
B_T	1	1
age of τ_4 at T	9	9

Under D1, W_T is 1 with or without the stuck task, so W_T alone cannot see it. The run is still not hidden. τ_4 adds one to the backlog B_T, which is weighted in $\mathcal{J}$, and its age of 9 timesteps is reported with the unfinished tasks.

5.2 The two options

D1. Keep W_T as it is, and say why. Every per-task average in the section (service time, movement cost, blocked time) is taken over the same completed tasks, so they can be compared directly. Unfinished work is the job of one quantity, the backlog. This needs one sentence in the thesis and no model change.

D2. Count every assigned task. W_T also sees the stuck task. The cost is that W_T is then averaged over a different set of tasks than $\bar{\zeta}_T$ and $\bar{c}_T$, and a task that is stuck counts twice in $\mathcal{J}$, once through W_T and once through B_T. This is a model change.

Recommendation. D1. It is the same rule service time already follows, since $\bar{\zeta}_T$ also ignores τ_4. The worked example shows that the backlog catches the stuck task. D2 would mostly punish the same task a second time.

Your decision. D1 or D2?

6 Decision E. The name "movement cost"

Agreed. E1. Keep the name and add one clause saying that standing still costs c_{wait} per timestep, so $\bar{c}_T$ counts it too.

7 New structure of the section

Agreed. The structure below. The waiting group now also holds the crowding measure $\bar{\delta}_T$ from Decision B.

The reader first gets a plain overview, then one group of quantities at a time under a `\paragraph` heading. The summary table moves to just after $\mathcal{J}$, where every symbol in it has been met.

1. **Overview**, with no symbols. A run answers three questions: how much work was done, what it cost, and how evenly traffic spread. One criterion then combines them.
2. **Completed work.** Throughput Λ_T and mean service time $\bar{\zeta}_T$, with the timeline figure (Section 8).
3. **Cost of the work.** $\bar{c}_T$.
4. **Unfinished work.** Backlog B_T.
5. **Waiting and crowding.** W_T on the same timeline figure, how it differs from $\hat{w}_t(e)$, then $\bar{\delta}_T$ with a pointer to `fig:congestion`.
6. **Traffic spread.** H_T and C_T with the new concentration figure, then its limits.
7. **The criterion.** $\mathcal{J}$, why throughput is left out, then the summary table.
8. **Runtime.** κ_T and why it stays out of $\mathcal{J}$.
9. **The decision problem.** `eq:decisionproblem`, the constraints, `eq:rqformal` and its factors, and robustness (Decision C).

The per-task quantities (2 to 5) come before the per-edge ones (6), so the timeline figure serves four measures in a row. The line "Figure 12 shows the two extremes" disappears as a sentence of its own. The figure is referred to inside the paragraph that introduces C_T.

The proposed opening paragraph reads:

> A run is judged by three questions. How much work did the agents complete, what did that work cost, and how evenly did the traffic spread over the warehouse while they did it? Each question is answered by one or two numbers, defined below one group at a time. Five of them are then combined into a single score for the run, and the others are reported next to it.

8 Figures

8.1 Concentration figure (replaces `concentration.png`)

Agreed. The new figure replaces `concentration.png`.

Figure 1 is the draft. It is built on the same base as `observation.png`: the dimmed warehouse with vertex roles, D1 and D2, and agents a_1 to a_4 at the same positions ($m = 4$). Edge width and the printed numbers are traversal counts. The thesis version would keep two panels, or three under A1, since the third panel is the case that motivates A1. The two normalisations are printed only in this draft.

The text example of three edges used 8, 1 and 1 times would be replaced by the figure's numbers: 72 traversals, the eight edges from a_4 to D1 used six times each, and the D2 aisle unused.

8.2 Timeline figure (new, replaces `tab:throughputexample`)

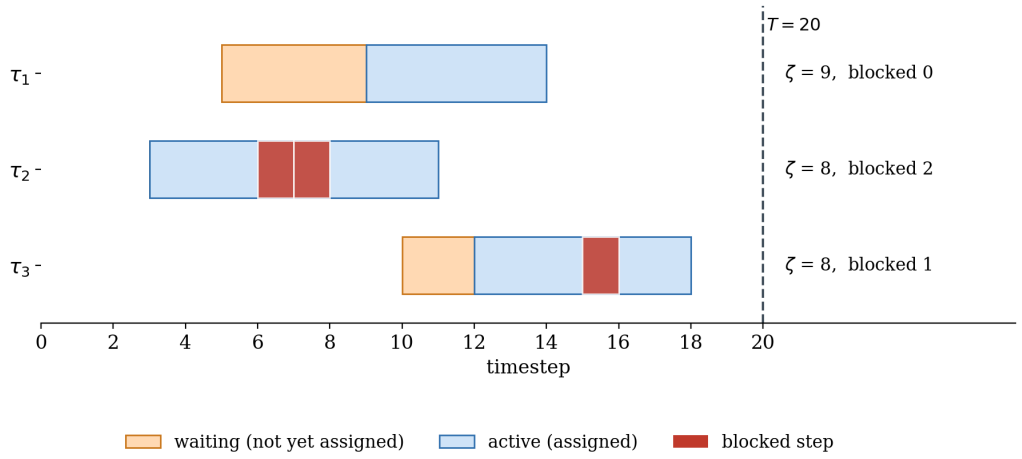


Figure 3: Draft timeline of the three tasks of the toy table, with blocked steps. It gives $\bar{\zeta}_T \approx 8.33$, $\Lambda_T = 0.15$ and $W_T = (0 + 2 + 1)/3 = 1$.

You asked for the blocked steps to be added to `lifecycle.png`. That does not work without contradicting the task example. In `fig:taskroute`, a_2 moves from its start at $t = 9$ to A at $t = 11$ and to D1 at $t = 14$, one edge per timestep. It is never blocked, so τ_1 contributes 0 to W_T.

The draft instead draws the three tasks of the toy table on one time axis and gives τ_2 and τ_3 a few blocked steps. One figure then shows service time, throughput and blocked time at once. It can replace the toy table, whose numbers it contains. `lifecycle.png` stays as it is.

Agreed. The timeline figure is added.

Your decision. You wrote "keep 6.2". I read that as adding the timeline figure and dropping the toy table, since the figure holds its numbers. If you meant that the toy table should stay next to the figure, say so.

8.3 Crowding figure

No new figure is needed. The crowding measure $\bar{\delta}_T$ points to `fig:congestion`, which already shows how $\delta_i(t)$ is read for each agent.

9 What will be applied once you decide

1. Restructure the section as in Section 7.
2. Fixes P1, P2, P3, P6, P9, P10 and P11, which need no model decision.

3. Decisions A to E as chosen (B and E as above), each model change with an `[Axx]` tag, a `GAPS.md` entry and a notation table update.
4. Add the figure scripts to `thesis-progress/scripts/images/`, with outputs copied to both `Images/` folders.
5. Build the thesis, check `git diff main.tex` for stray text, and commit only after you have read the result.